INCH-POUND

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DETAIL SPECIFICATION

GLOVES, TOXICOLOGICAL AGENTS PROTECTIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers impermeable butyl rubber gloves for protection of the hands against toxicological agents. The gloves are used in conjunction with impermeable protective clothing for the protection of personnel against toxicological warfare agents. The Type II gloves are to be worn with M-3 Coverall or M-2 Apron, Toxicological Agent Protective (Army) and the Type IIA gloves are to be worn in conjunction with the M-4 Coveralls in chemical plants for protection of personnel. This is a special purpose Life Support Clothing and Equipment (LSC&E) item.

1.2 Classification. This specification covers the following types and sizes of gloves as specified (see 6.2)

1.2.1 Types.

Type I - DELETED
Type II - Protective M2 and M3, without closure ring
Type IIA - Protective impermeable M4, without closure ring
Type III - DELETED
Type IV - DELETED

Comments, suggestions, or questions on this document should be addressed to: Attn: DLA Troop Support Standardization Team, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil/.

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1.2.2 Schedule of sizes.

Extra small (8) 1/
Small (9)
Medium (10)
Large (11)
Extra large (12)

1/ Not applicable to Type IIA.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-12468 - Decontamination Agent, STB
MIL-DTL-32066 - Gloves, Rubber, Industrial

(Copies of these documents are available online at https://quicksearch.dla.mil.)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-282 - Filter Units, Protective Clothing, Gas-Mask components and Relate Products: Performance Test Methods

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

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DRAWINGS

U.S. ARMY COMBAT CAPABILITIES DEVELOPMENT COMMAND SOLDIER CENTER

8-1-307 – Forms for Gloves, Chemical Protective

(Copies of drawings are available from the U.S. Army Combat Capabilities Development Command (CCDC), Soldier Center, ATTN: FCDD-SCP-WC, 10 General Greene Avenue, Natick, MA 01760-5019.)

U.S. ARMY COMBAT CAPABILITIES DEVELOPMENT COMMAND CHEMICAL BIOLOGICAL CENTER

C5-9-155 – Protective Equipment Suit, Protective Impermeable Cuff (Pants and Sleeve)

(Copies are available from U.S. Army Combat Capabilities Development Command (CCDC) Chemical Biological Center, Aberdeen Proving Ground, MD 21020-4223)

DEPARTMENT OF THE ARMY PUBLISHING DIRECTORATE

FM 3-11 - Chemical Biological, Radiological and Nuclear Operations

(Copies of this document are available online https://armypubs.army.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D297 - Standard Test Methods for Rubber Products - Chemical Analysis
ASTM D412 - Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers - Tension
ASTM D573 - Standard Test Method for Rubber-Deterioration in an Air Oven
ASTM D1053 - Standard Test Methods for Rubber Property – Stiffening at Low Temperatures: Flexible Polymers and Coated Fabrics
ASTM D3767 - Standard Practice for Rubber - Measurement of Dimensions
ASTM E1229 - Standard Specification for Calcium Hypochlorite

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(Copies of these documents are available online at https://www.astm.org.)

INFORMA HEALTHCARE

Repeat Insult Patch Test - Modified Draize Procedure –
Principles and Methods of Toxicology, A Wallace Hayes (editor).

(Copies of this document are available online at https://www.crcpress.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Unless a specific exemption has been obtained, nothing in this document, supersedes applicable laws and regulations.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Guide sample. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variations from this specification may appear in the sample in which case this specification shall govern.

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Design. The butyl rubber gloves shall be five finger style with rotated thumb made on glove forms conforming to the dimensions shown on Drawing 8-1-307 using a dipping process to produce the gloves of the required size. Type IIA may be made on standard commercial glove form. This is a special purpose Life Support Clothing and Equipment (LSC&E) item. All Government administrative and surveillance procedures applicable to LSC&E items shall be envoked in accordance with the contract or purchase order (see 6.2).

3.5 Material. The gloves shall be made of butyl rubber, pigmented black, suitably compounded and vulcanized to meet the requirements specified in Table I when tested as specified in 4.5.

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TABLE I. Physical and chemical requirements of rubber gloves.

Characteristic	Requirement
Thickness, inch	0.035 (±0.010)
Tensile strength, p.s.i., (min.):	
Original	1100
After aging	1000
After decontamination solution immersion	800
Tensile strength at 200 percent elongation, p.s.i.	325 (±125)
Ultimate elongation, percent, (min.):	
Original	400
After aging	350
After decontamination	300
Total bound sulfur, percent, (min.)	65
Low temperature flexibility (Type II)	Pass 1/
Mustard resistance, minutes, (min.)	360
GB resistance, minutes, (min.)	450
Deformation due to contamination solution, (max.):	
After immersion wet, percent increase in thickness	10.0
After immersion wet, percent increase in length	15.0
After immersion and aeration, percent increase in thickness	6.0
After immersion and aeration, percent increase in length	4.0
Porosity	Pass 2/

1/ The rubber specimens shall have an angular twist no less than the values shown below for the thickness indicated. Interpolation shall be used for any thickness not shown below:

Thickness, inch	Twist, angular degrees, (min.)
0.025	138
0.030	119
0.035	99
0.040	84
0.045	70

2/ There shall be no sign of porosity.

3.6 Color and finish. The outside of the finished gloves shall be black with a dull finish.

3.7 Construction.

3.7.1 Types II. The Types II gloves shall be finished with a rolled edge around the gauntlet no greater than 1/4-inch thick.

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3.7.2 Type IIA. The diameter of the glove gauntlet shall be dimensionally compatible with the outside diameter of the plastic sleeve cuff specified in the U.S. Army Combat Capabilities Development Command (CCDC) Chemical Biological Center Drawing C5-9-155. When the plastic cuff, or suitable fixture of the same exterior dimensions as the cuff (see 6.2) is inserted into the glove so the large diameter of the plastic cuff is flush with the top of the glove, the entire exterior surface of the cuff or fixture shall be in contact with the glove gauntlet. The glove shall be finished with the trimmed straight edge around the top of the gauntlet.

3.7.3 Dusting. The inside and the outside of the finished glove shall be lightly dusted with whiting, talc, or other finely divided non-irritant, non-sensitizing, non-toxic material powder which does not support mildew growth.

3.8 Dimensions.

3.8.1 Length. The length of Type II gloves shall be 14 (± 1/2) inches. The length of the Type IIA gloves shall be 16 (± 1/2) inches. The overall length of the Types II and IIA gloves shall be measured from the tip of the middle finger to the cuff edge with the glove flattened and unstretched.

3.8.2 Width. The width of the gloves shall be as specified in Table II. The glove shall be flattened without stretching and the width measured across the palm at the thumb crotch on a line parallel to the cuff edge.

TABLE II. Glove dimensions.

Size	Width (inches)all Type II Tolerance (±1/8) inch	Width (inches)all Type IIA Tolerance (±1/4) inch
Extra small (8)	4-1/8	-
Small (9)	4-5/16	4-5/8
Medium (10)	4-1/2	5-1/8
Large (11)	4-13/16	5-5/8
Extra Large (12)	5-1/8	6-1/8

3.9 Identification marking. Each glove shall be permanently and legibly marked in bold-faced Gothic capital letters of not less than 1/4 inch in height with either yellow or white indelible marking ink (see 6.5). The palm side of the all gloves shall be marked with the glove size and the following legend: (Marking shall be centered approximately one (1) inch from the cuff edge.)

"DO NOT USE FOR ELECTRICAL WORK OR FIRE FIGHTING"

The back side of Types II gloves shall be marked with the following information:

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"GLOVES, TOXICOLOGICAL AGENTS PROTECTIVE"

The back side of the Type IIA gloves shall be marked with the following information:

"GLOVES, PROTECTIVE, IMPERMEABLE, M 4"

The back side of all gloves shall be marked with the following information:

SPECIFICATION NO.
CONTRACT NO.
NATIONAL STOCK NO.
TYPE

3.10 Instruction sheet. A paper instruction sheet shall be placed inside one glove of each pair of gloves. The size and style of the letters shall be 1/4 inch (18 point) for heading and 1/8 inch (10 point) for instructions on the use and decontamination of gloves with all lettering legible printed in black Demibold Futura capital letters. The instructions sheet shall contain the following information:

1. If during use, the gloves come in contact with toxicological agents, decontaminate and wash them with water immediately to avoid contaminating other surfaces.

2. If gloves develop holes, become torn, cracked or damaged before use or during use, discard gloves immediately and replace with a serviceable pair.

DECONTAMINATION OF GLOVES

Gloves exposed to toxicological agents must be decontaminated before reuse. The gloves can be decontaminated by using the procedures specified in FM 3-11, Chemical Biological, Radiological and Nuclear Operations.

3.11 Resistance of marking to decontamination procedure. The marking on the glove shall remain clear and legible when tested as specified in 4.5.

3.12 Workmanship. The finished gloves shall be uniform in quality and shall conform to the quality of product established by this specification. The occurrence of defects, as specified in 4.3, shall not exceed the criteria specified in the contract or purchase order.

3.13 Toxicity. The finished glove shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.5. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article inspection. The first article, submitted in accordance with 3.1 shall be inspected for compliance with design, workmanship and dimensional requirements. The presence of excessive defects, as defined by contract (see 6.2) or failure to pass any test shall be cause for rejection of the first article. The sample size shall be specified in the procurement document (see 6.2). The sample unit shall be one pair of glove and the lot size shall be expressed in pairs of gloves.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4, 4.4.1 and 4.4.2 and the tests of 4.5 through 4.7 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits as specified in the contract and/or order, except where otherwise indicated.(see 6.2).

4.3.1 Inspection conditions. Unless otherwise specified, excluded, amended, modified or qualified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with all the requirements of referenced documents.

4.4 End item critical defect examination. Every glove, prior to visual and dimensional examinations in 4.4.1 and 4.4.2 shall be examined for the critical defects listed in Table III. Any end item found to contain one or more critical defects shall be rejected.

4.4.1 End item visual examination. The gloves shall be examined for the defects listed in Table III. The lot size shall be expressed in units of gloves. The sample unit shall be one glove and the selection shall be by pairs. Defects for pairing shall be classified as a single defect. The inspection level shall be as specified in the contract/ and or purchase order. The finding of one or more critical defect shall be cause for rejection of the lot.

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TABLE III. End item visual defects.

Examine	Defect	Classification		
		Critical	Major	Minor
Pairing	Not properly mated, e.g., not right and left of the same size. Definite variation in color and appearance.		101	 201
Design	Any characteristic not in accordance with specified requirements (unless otherwise indicated herein).		102	
Color and finish	Not black Not dull finish			202 203
Construction and workmanship: General	Any cut, tear, hole, rip, or rupture through material. Any closed blister. Any burned sport which cracks on flexing, bending, or stretching (by hand). Any pinching, pit, thin sport, abraded area, deep crease, readily removable foreign matter (see NOTE below), or similar defective condition which results in a thickness at the defect less than the minimum thickness allowed for the glove.	1 2 3 4		
	Any one (1) inch square area which contains more than three (3) pinches, pits, thin spots, abraded areas, deep creases, readily removable foreign matter, (see NOTE below), or similar defective condition not resulting in a thickness allowed for the glove. Any foreign matter not readily removable in area beyond 4-inches from cuff edge. Any foreign matter larger than 1/23 inch across, not readily removable, in area within 4-inches from cuff edge.			204 205 206
NOTE:	Readily removable foreign matter is defined as material which can be removed by flexing or stretching the glove by hand or by rubbing the defective area with the ball of the thumb or fingers. Cutting, scratching, or otherwise puncturing the surface is not permitted.			

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TABLE III. End item visual defects. - Continued

Examine	Defect	Classification		
		Critical	Major	Minor
Construction and workmanship: General (cont'd)	Any repair or patch.		103	
	Any solid rubber ridge, run or lump resulting in a thickness greater than the maximum thickness allowed for the glove.			207
	Tackiness after dusting (see 3.7.3).	5		
	Any malformation or distortion seriously affecting serviceability.		104	
	Any malformation or distortion affecting serviceability but not seriously.			208
	Not clean, i.e. dirty but can be cleaned with a cleaning agent.			209
	Not clean, i.e. dirty but cannot be cleaned with a cleaning agent.		105	
	Inside and outside of glove not dusted.		106	
Type II	Edge not rolled as specified.		107	
	Rolled edge more than 1/4-inch thick			210
	Roll crooked or loose			211
Type IIA	Gauntlet of glove not dimensionally compatible with exterior of plastic cuff.		108	
	Top of gauntlet not finished with trimmed straight edge.			212
Marking, identification, and instruction sheet	Not permanent, i.e., can be easily rubbed off with moistened thumb.			213
	Not in specified location, characters not of height specified, not specified color or not accomplished as specified.			214
	Omitted or illegible.		109	
	Incomplete or incorrect but not to the extent that the intended purpose will not be served.		110	
	Incomplete or incorrect but not to the extent that the intended purpose will not be served.			215

4.4.2 End item dimensional examination. The gloves shall be examined for conformance to the dimensions as specified in 3.8. Any dimension not within the specified tolerance shall be classified as a defect.

4.5 End item testing. The gloves shall be tested for the characteristics listed in Table IV. The methods of testing as specified wherever applicable and as listed in Table IV shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be pairs 9 pairs of gloves for all physical and chemical tests. All

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test reports shall contain the individual values used in expressing the final results. The lot shall be unacceptable if one or more sample units fails to meet any requirement specified. The production lot sample size shall be in accordance with the following:

Lot size (pairs of gloves)	Sample size (sample units) (pairs)
400 or less	18
401 up to and including 11,000	27
11,001and over	45

a. Government chemical agent tests (see 6.6). The Government shall perform acceptance tests for mustard and GB resistance. The sample unit shall be two pairs of gloves and the inspection level shall be as specified in the contract and/or purchase order. These tests are considered critical and any failure shall be cause for rejection of the lot represented by the sample. The sample size shall be as specified in the contract and/or purchase order.

b. Porosity test. The maximum lot size for the porosity shall be 20,000 pair of gloves. The sample unit shall be one glove and the sample size shall be 800 individual gloves. Any glove found to leak shall be cause for rejection of the lot represented by the sample. The sample size shall be as specified in the contract and/or purchase order.

TABLE IV. End item tests.

Characteristic	Requirement	Test method	Number of determinations per sample unit	Results reported as	
				Pass or fail	Numerically to nearest
Material identification (rubber)	3.4.1	ASTM D297 1/	1	X	
Thickness	Table I	4.6.1	-	-	
Porosity	Table I	4.6.2	1	X	-
Low temperature flexibility (types II)	Table I	4.6.3	-	X	-
Deformation due to contamination solution:	Table I				
After immersion wet:					
Thickness percent		4.6.4	-	-	0.1 percent
Length percent		4.6.4	-	-	0.1 percent
After immersion and aeration:					
Thickness percent		4.6.4	-	-	0.1 percent
Length percent		4.6.4	-	-	0.1 percent

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TABLE IV. End item tests. - Continued

Characteristic	Requirement	Test method	Number of determinations per sample unit	Results reported as	
				Pass or fail	Numerically to nearest
Tensile stress at 200 percent elongation	Table I	ASTM D412 2/	-	-	-
Ultimate elongation: Original After aging After decontamination solution immersion	Table I	ASTM D412 2/ ASTM D412 2/ 3/ 4.6.5	-	-	-
Mustard resistance	Table I	T209 4/	2	-	1 minute
GB resistance	Table I	T208 4/	2	-	1 minute
Total bound sulfur	Table I	ASTM D297 and 4.6.6	3	-	0.1
Resistance of marking to decontamination procedure	3.11	4.6.7	1	X	-
Toxicity	312	4.7			

1/ Only pyrolysis test shall be used.
2/ Three gloves shall be selected from each sample unit and one specimen shall be tested from each glove selected. Method A of the test shall be used.
3/ The specimens shall be aged at a temperature of 212ºF for 48 hours in accordance with ASTM D573 before being tested for tensile strength and elongation.
4/ Refers to test method in MIL-STD-282.

4.6 Methods of inspection.

4.6.1 Thickness test. Thickness of the gloves shall be determined in accordance with ASTM D3767 using a micrometer graduated to 0.001 inch, having an anvil about ¼ inch in diameter and a circular foot of 0.124 to 0.126 inch in diameter, exerting a total force of 2.6 ounces. Five measurements shall be made on each of the three gloves in the area not less than 1-inch nor more than 5-inches from the cuff edge. The crotches of the gloves shall also be measured for thickness. The sample unit shall fail if any single measurement is less than the minimum or greater than the maximum specified.

4.6.2 Porosity test. The porosity shall be performed as specified in MIL-DTL-32066 except that the air pressure shall be 0.5 pounds per square inch and the inflated glove shall be immersed in the water in stages, for observation of porosity as indicated by air bubbles. The inflated gloves shall be immersed in water, cuff or fingers first, in three stages. Initially, the glove shall be immersed so that about 1/3 of the glove in under water. The glove will be carefully observed

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for the release of air bubbles. If no air bubbles are noted, the glove shall be lowered until 2/3 of the glove is under water. Again carefully observe the area for the presence of air bubbles. If no air bubbles are observed the reminder of the glove shall be immersed in the water so that the tip of the middle finger or cuff bead is less than 1 inch below the water line. Carefully observe the glove for air bubbles. No visible bubbling shall be permitted during the test. At no time should the area of the globe being observed for air bubbles be further than 6 inches below the surface of the water. The pressure exerted by the water below this 6 inch depth can be great enough to prevent the release of air bubbles from a small hole or tear.

4.6.3 Low temperature flexibility test (Types II). Low temperature flexibility shall be determined in accordance with ASTM D1053, paragraph entitled Routine Inspection and Acceptance with the following exceptions:

a. The test shall be conducted at -20°F.
b. The liquid coolant shall be methyl alcohol.
c. The exposure time shall be 5 minutes.
d. The black wire having a torsional constant of 0.125 gf.cm/degree shall be used instead of the standard wire.
e. The angular degree of twist shall be in accordance with Table I.

4.6.4 Deformation due to decontamination solution test. Three gloves from each sample unit shall be tested. The resistance to decontamination solution shall be determined as follows:

a. Accurately mark a three (3) by six (6) inch area on each glove. Draw three (3) straight lines along the entire length of the area as follows: The first line shall be 1/2 inch from one edge of the marked area, the second line 1/2 inch in from the opposite edge. The third line shall be 1-1/2 inches from both edges. Then draw three straight lines across the entire width of the area as follows: The first line shall be two (2) inches in from one edge of the marked area, the second line two (2) inches in from the opposite edge. The third line shall be three (3) inches from both edges. Using a micrometer as described in ASTM D3767 determine the thickness of the area at the nine points where the three length lines cross the three (3) width lines. The thickness shall be the average of the nine (9) values.

b. Mount the glove on a suitable form and immerse in test solution to within one (1) inch of the cuff edge, taking care not to wet the interior. The test solution shall be a Super-Tropical Bleach slurry. The slurry shall be made up (by weight) of 1-part Super-Tropical Bleach (STB) conforming to MIL-DTL-12468 and 2 parts water. (NOTE: Super-Tropical Bleach is corrosive to most metals and is damaging to most fabrics.) The temperature of the slurry during the test shall be 80°F to 90°F. The time of immersion shall be five (5) minutes.

c. After the five (5) minute immersion in STB slurry, withdraw the glove, wash off the slurry with water and rinse in 95 percent ethyl alcohol. While the glove is still wet, measure the marked area immediately for thickness and length. Thickness shall be determined as described in “a” above. The length shall be measured along the three (3) length lines and the length recorded as the average of the three (3) determinations.

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d. Place the wet gloves on paper towels that are heavily covered with talcum powder (with marked area facing the talced paper towel) and allow to stand at room temperature for 18 hours.

e. After the 18 hours air drying, re-measure the marked area for thickness and length. Determine the thickness in accordance with "a" above. Determine the length in accordance with "c" above.

f. The percent increase in thickness and percent increase in length shall be calculated as follows:

$$\text{Percent increase in thickness after immersion (wet)} = \frac{T1 - T}{T} \times 100$$

$$\text{Percent increase in thickness after immersion and aeration} = \frac{T2 - T}{T} \times 100$$

$$\text{Percent increase in length after immersion (wet)} = \frac{L1 - L}{L} \times 100$$

$$\text{Percent increase in length after immersion and aeration} = \frac{L2 - L}{L} \times 100$$

Where:
T = Original thickness as determined in "a" above.
T1 = Thickness after immersion as determined in "c" above.
T2 = Thickness after immersion and aeration as determined in "e" above.
L = Original length (6 inches)
L1 = Length after immersion as determined in "c" above.
L2 = Length after immersion and aeration as determined in "e" above.

4.6.5 Tensile strength and ultimate elongation after decontamination test. The tensile strength shall be determined in accordance with ASTM D412 (Method A). The cross sectional area of the specimen shall be calculated using the thickness determined in 4.6.4a. The ultimate elongation shall be determined in accordance with ASTM D412 (Method A). One (1) specimen shall be cut from each glove that was tested for decontamination. The specimen shall be cut from the three (3) by six (6) inch area marked on the gloves. The long dimension of the three (3) by six (6) inch area. The specimen shall be cut in such a manner that each end of the specimen is the same distance from the edge of the marked area.

4.6.6 Total bound sulfur. The total bound sulfur shall be determined in accordance with ASTM D297. One (1) glove from each sample unit shall be tested. Three (3) individual readings shall be taken on each of the gloves. The sample unit shall fail if any single measurement is less than the minimum specified in Table IV. The percent of the total bound sulfur shall be calculated as follows:

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$$\text{Total bound sulfur} = \frac{(TS-FS)}{TS} \times 100$$

Where:

TS = Percent total sulfur
FS = Percent free sulfur

4.6.7 Resistance of marking to decontamination procedure. The gloves shall be tested as follows:

a. Boil gloves for 2-1/2 to 3 hours in a solution containing 1.25 ounces per gallon of calcium hypochlorite conforming to ASTM E1229.

b. Remove gloves from the bleach solution and soak in fresh water at 70°F for 15 minutes. Change the water and repeat this step two (2) additional times.

c. Hang the gloves and allow them to dry at about 110°F.

d. When dry, examine all types of gloves for legibility of marking.

4.7. Toxicity test. When required, (see 6.2), an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of these studies indicate the glove is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 2.3). If the toxicity requirement (see 3.12) can be demonstrated with historical use data on the finishing treatments used, toxicity testing may not be required (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The gloves are intended to be used in conjunction with impermeable protective clothing for the protection of personnel against toxicological warfare agents. The Type II gloves are to be worn with M-3 Coverall or M-2 Apron, Toxicological Agents Protective (Army). The Type IIA Gloves are to be worn in conjunction with the M-4 coveralls in chemical plants for protection of personnel.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and Sizes required (see 1.2)
c. The specific issue of individual documents referenced (see 2.2).
d. When first article is required (see 3.1, 4.2, 6.3).
e. Inspection conditions (see 4.1.1)
f. Conformance inspection acceptance quality limits (AQL) (see 4.3)
g. When toxicity testing is required (see 4.7)
h. Packaging (see 5.1).

6.3 First article. When a first article inspection is required (see 3.1), it will be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Guide samples. For access to guide samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Marking ink. Markem JXN-7410-H White or Yellow made by Markem Machine Co., 40 Putnam Street, Keene, NH, has been found to be suitable when dried under infrared light for 24 hours.

6.6 Tests. Items for testing of HD resistance, GB resistance will be directed to JEFS Test Management Services ATTN: Cathy Campbell Joint Testing Operations Center 120 Maple Street London, OH 43140

6.7 Subject term (key word) listing

Butyl rubber
Five finger
Impermeable

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians:
Army - GL
Navy - NU
Air Force - 11

Preparing Activity:
DLA-CT

Review activities:
Army - EA, MD

(Project: 8415-2020-035)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil/.

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